RWANDA POLYTECHNIC

NGOMA COLLEGE

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SEMESTER I

LEVEL 7

MODULE TITLE: WORKPLACE ENVIRONMENTAL MANAGEMENT

MODULE CODE: CCMEM701

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Trainer: MPABUKA Shaban

TEL: +250788210137

Purpose statement

This module describes the knowledge, skills, and attitudes required to manage workplace environmental management. Throughout this module, learners will be able to describe environmental management, assess the environmental impact, and control human projects' impact on the environment.

Learning units

Learning unit 1 - Describe environmental management

Learning unit 2 - Assess the environmental impact

Learning unit 3 - Control human projects' impact on the environment

LEARNING UNIT 1 - DESCRIBE ENVIRONMENTAL MANAGEMENT

Learning Outcomes:

1.1 Explain the basics of environmental management

1.2 Describe the environment of the earth

1.3 Describe Rwanda's environmental features

LEARNING OUTCOME 1.1: EXPLAIN THE BASICS OF ENVIRONMENTAL MANAGEMENT

1.1.1. ENVIRONMENTAL MANAGEMENT KEY TERMS

- **Environment:** Environment is a set of physical, chemical, and biological elements as well as socio-economic, cultural, aesthetic, and intellectual factors likely to have a direct or indirect, immediate or long-term impact on the development of the environment, human beings, and human activities.
- **Management:** Management is a set of principles relating to the functions of planning, organizing, directing, and controlling, and the application of these principles in harnessing physical, financial, human, and informational resources efficiently and effectively to achieve organizational goals.
- **Environmental management:** Environmental Management means a process to efficiently and systematically consume the environment by planning, operating, monitoring, improving, and enhancing, which concerns saving, highest utilizing, sustainability, longlasting, and providing the most beneficial to humans and nature.
- **Environmental policy:** any measure by a government or corporation or other public or private organization regarding the effects of human activities on the environment, particularly those measures that are designed to prevent or reduce harmful effects of human activities on ***ecosystems***.

  An ecosystem is a term that is commonly used in biology and environmental studies. In general, it is defined as a system or community that includes all living organisms and their non-living components in nature. All of these components work together, either directly or indirectly, to maintain ecological balance.

  International environmental policy: covers a number of issues: climate protection, sustainable energy policy, preservation of biological diversity, and the conservation of forests, seas, and soils. Further related topics are desertification, sustainable waste management, and protection against hazardous substances.

  The international standard ISO 14001 is a standard for management that aims to support comprehensive environmental protection.

The standard is written in a manner that is applicable to all types and sizes of organizations and can be adapted to different geographical, cultural, and social conditions.

1.1.2. CHARACTERISTICS OF ENVIRONMENTAL MANAGEMENT

1. **It deals with a world affected by humans;** Humans impact the physical environment in many ways: overpopulation, pollution, burning fossil fuels, and deforestation. Changes like these have triggered climate change, soil erosion, poor air quality, and undrinkable water. These negative impacts can affect human behavior and can prompt mass migrations or battles over clean water.
2. **It supports sustainable development;**

It improves the quality of human life,

It minimizes the depletion of natural resources,

It teaches us to respect and care for all life forms,

Checking the pollution levels,

Making arrangements so that the future generation is able to meet their own demands

3. **It demands a multidisciplinary approach;**
4. **It has to integrate different development viewpoints;**
5. **It concerns with short-term and long-term planning as well as from local to global scale; and**
6. **It seeks to integrate natural and social science, policy-making, and planning.**

- **Purpose of environmental management:** An EMS helps your organization achieve environmental goals and systematically address regulatory requirements in a cost-effective manner. It provides a proactive approach to working towards environmental targets, reducing risk, and improving health and safety practices. Better yet, an Environmental Management System can help your company address and work towards non-regulated issues that are more important globally, like energy conservation.
- **Advantages to undertaking Environmental management:**

Most advantages of an EMS system from savings your business can make, increased profitability, and better sales opportunities:

- Better regulatory compliance - running an EMS will help ensure your environmental legal responsibilities are met and more easily managed on a day-to-day basis.

- More effective use of resources - you will have policies and procedures in place that help you manage waste and resources more effectively and reduce costs.
- Marketing - you can highlight your business' credentials as an environmentally aware operation that has **made a commitment to continual environmental improvement through advertising or annual reporting**.
- Finance - you may find it easier to raise investment from banks and other financial institutions, which are increasingly keen to see businesses controlling their environmental impact.
- Increased sales opportunities - large businesses and **government departments may only deal with businesses that have an EMS**.
- Lighter regulation - even if an EMS is not a regulatory requirement, by showing your commitment to environmental management, you may benefit through less frequent site visits or reduced fees from environmental regulators.
- Certification to recognized standards - gaining external certification of your EMS through ISO 14001, BS 8555 or EMAS can **give your business credibility with customers and stakeholders**.

1.1.3. TYPES OF ENVIRONMENTS

- ✓ **Natural environment**: The natural environment consists of land-based ecosystems such as grasslands and forests, aquatic ecosystems such as rivers and wetlands, and coastal and marine ecosystems such as mangroves and sea-grass meadows. Ecosystems include animals and plants and the interactions they have with each other and their physical environment. Climatic conditions, particularly rainfall patterns and fire regimes, strongly affect the type and extent of an ecosystem and the animals and plants that are present.

Note that Land, air, water, plants and animals all comprise the natural environment. Let us learn about the different domains of the natural environment. These are found in these four components which are **lithosphere, hydrosphere, atmosphere and biosphere**.

- ✓ **Man-made or Built environment**: The built environment touches all aspects of our lives, encompassing the buildings we live in, the distribution systems that provide us with water

and electricity, and the roads, bridges, and transportation systems we use to get from place to place.

Apart from the above, the other 7 components of the built environment affect human health: **air quality, ambient noise, climate, land use and development patterns, physical activity, public spaces, and transportation**. Each one has a significant impact on our well-being.

Most industrial environment which are part of built environment are: warehouses, plants, manufacturing, or fabrication facilities. These industrial environments are often using the advanced heating and cooling controls (HVAC), and are sometimes partially exposed to the elements. Industrial environments may also increase equipment exposure to dust and other contaminants.

- ✓ **Social environment:** A person's social environment is their society and all surroundings are influenced in some way by humans. It includes all relationships, institutions, religions, culture, leisure, and physical structures.

Also, the social environment includes **the groups to which we belong, the neighborhoods in which we live, the organization of our workplaces, and the policies we create to order our lives**.

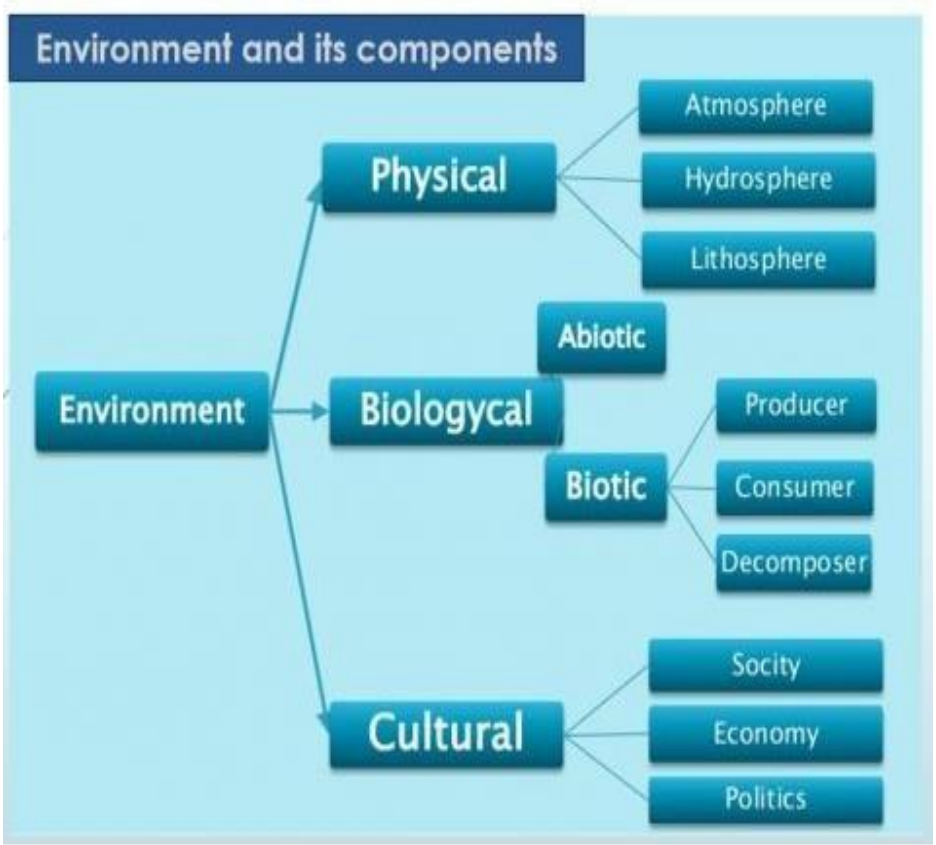
The natural environment is the natural world around us: the ground, the trees, and the air. The social environment is, collectively, all of the things that humans have overlaid on top of our world: our personal and societal relationships, our institutions, our cultures, and our physical surroundings all of the aspects and products of human activity and interaction.

1.1.4. COMPONENTS OF THE ENVIRONMENT

- **Physical elements**: The physical component refers to the non-living part of the environment. These are also called abiotic factors and include air, water, soil, climate, etc. Physical components can be broadly classified into 3 groups – Atmosphere, Hydrosphere, and Lithosphere. Additionally, scientists also use the term "biosphere" to describe the zone of life (or the worldwide sum of ecosystems).
- **Biological elements**: The biological component of the environment, as the name indicates, includes all living things. Hence, it is also called the biotic component of the ecosystem. Animals, plants, and microorganisms interact with the abiotic components, and these form

various ecosystems. Moreover, organisms in these ecosystems are classified into groups – such as producers, consumers, and decomposers. They also occupy various trophic levels on the food chain. For instance, all autotrophs are producers and occupy the lowest levels on a food chain.

- **Cultural/social elements**: Cultural environments are environments shaped by human activities, such as cultural landscapes in the countryside, forests, urban areas, and cities, fixed archaeological structures on land or water, constructions and built environments from different ages, along with bridges, roads, power lines and industrial and harbor areas. On the one hand, cultural environments are a non-renewable resource, on the other hand, they are in a constant state of renewal and development.



1.1.5. TYPES OF ENVIRONMENTAL MANAGEMENT

- **The construction environmental management plan**: Construction environmental management plans often detail the implementation of measures in accordance with environmental commitments outlined in; an Environmental Statement, an environmental policy or environmental plan, requirements of planning conditions, Section 106 agreements, or other legislative requirements. They are 'live' documents that are reviewed and updated at regular intervals throughout the project life cycle.
- **The operations environmental management plan:** An Operations Environmental Management Plan (OEMP) provides specific guidance related to the operational activities associated with the facility. The roles and responsibilities for mitigation, monitoring, and performance assessment during the facility's operational life are also specified. The OEMP should not become an additional requirement separate from the day-to-day activities of the site, to avoid it being seen by operations staff as an obstruction to their normal duties and operations, hence it must be part of a facility's routine operations and
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In new operational facilities, the OEMP is usually prepared using the framework defined by the project EIA Study and/or the CTO. When an EIA Study is not available, the result from environmental assessments conducted during earlier stages of the project, as well as updated

information on the project as per the detailed engineering design implemented, would be used as a starting point when preparing the OEMP.

For projects within existing operational areas, the proposed OEMP becomes a part of the existing facility's OEMP.

✓ **Objective of OEMP**

The key objectives of OEMP are summarised as follows:

- Compliance with environmental law and project commitments
- Avoidance, minimization, and or mitigation of environmental impacts
- Facilitate the timely distribution of information and increase awareness
- Manage assets in accordance with design criteria
- Provide a guide to the management and notification of environmental incidents
- Effective response to community requests and complaints; and
- General improvements in environmental maintenance methods.

✓ **Components of the OEMP**

Environmental concerns defining this OEMP are unique to the project's operational phase.

- Environmental issue
- Environmental objectives
- Environmental intent
- Control measures
- Responsibility
- Monitoring
- Reporting
- Performance Indicators
- Corrective Action

- **The decommissioning environmental management plan:** Decommissioning Plan means a plan to retire the physical facilities of the Project, including decontamination, dismantlement, rehabilitation, landscaping, and monitoring.

1.1.6. PRINCIPLES OF ENVIRONMENTAL MANAGEMENT

- ✓ **Polluter Pays Principle (PPP)**: The 'polluter pays' principle is the commonly accepted practice that those who produce pollution should bear the costs of managing it to prevent damage to human health or the environment. For instance, a factory that produces a potentially poisonous substance as a by-product of its activities is usually held responsible for its safe disposal. The polluter pays principle is part of a set of broader principles to guide sustainable development worldwide (formally known as the 1992 Rio Declaration).
- ✓ **The User Pays Principle (UPP):** This principle has been derived from the polluter pays principle which gives the responsibility to users for them to pay for any long-run cost and marginal environmental damage or pollution. It includes users bearing the costs for utilizing resources, services, and treatment services whenever the resources are consumed and used. For instance, consumption of water that comes from rivers, each household is required to pay a certain fee for the service. Farmers are required to pay land fees, of which part of the money goes towards cleaning and budgetary funding for developing EIA systems to help predict, protect, and prescribe measures to protect the environment from economic activities.
- ✓ **The Precautionary Principle (PP):** This proposes protecting the environment through precautionary measures, especially for heavy activities that might cause more damage to the environment. The precautionary principle has major objectives which include measuring primary and secondary activities posing a threat to the environment. Then industries are advised on ways and methods to implement which will not affect the environment so much throughout their daily activities. Precautionary activity measures the impact of any company and its activities, prescribing methods and measures with a less negative impact on its environment according to environmental impact assessments carried out at the time. The precautionary principle has been essential to protecting the environment, people, safe environment, implementation of policies, and reducing degradation and soil erosion

- ✓ **Principle of Effectiveness and Efficiency**: It is the responsibility of government in every country, city, or state to ensure, well-structured policies and procedures are put in place for essential waste management. Failure to properly mage waste can lead to diseases, soil problems, chemical build-ups, and water-borne diseases. Hence it's essential that through the principle of effectiveness and efficiency, major agencies and councils do everything possible to reduce waste building up and control dump sites for garbage. Encourages various organization bodies and agencies to decentralize, and implement new methods of management, proposed through new public management NPM to enable them to attain desired results when protecting the environment at a minimal cost.
- ✓ **The Principle of Responsibility:** Among principles of environmental management, this states each person and firm needs to be held accountable and take responsibility to maintain safe, clean, and sustainable development. Ecological sustainability should be attained by ensuring the use of resources is properly managed and not wasted, people must go about knowing that one of their duties is to protect the environment, safe applies for firms and corporations extracting and committing gasses polluting the environment.
- ✓ **The Principle of Participation:** Every citizen, person, government, and firm has a responsibility to participate in environmental decision-making and protection policies.

  Through collective collaboration in the affairs of the environment, it's easy to foster a shift and wave reflecting the need to protect the environment. Every individual should take a major step and contribute to issues relating to solid waste management, garbage collection, construction, chemicals, gaseous omission, and demolition materials that are likely to affect the environment and how to reduce the impact.
- ✓ **The principle of proportionality:** Refers to striking a balance between developments and protecting the environment. Building of basic essential infrastructure through development has been considered a major part of Human development, therefore, protecting the environment but so does development. Without the environment which provides for land, man would not have where to build homes.

1.1.7. ENVIRONMENTAL MANAGEMENT TOOLS

- ✓ **Environmental management systems:** An Environmental Management System (EMS) is a framework that helps an organization achieve its environmental goals through consistent review, evaluation, and improvement of its environmental performance. The assumption is that this consistent review and evaluation will identify opportunities for improving and implementing the environmental performance of the organization. The EMS itself does not dictate a level of environmental performance that must be achieved; each organization's EMS is tailored to its own individual objectives and targets.

What does an Environmental Management System do?

An EMS helps your organization achieve environmental goals and systematically address regulatory requirements in a cost-effective manner. It provides a proactive approach to working towards environmental targets, reducing risk, and improving health and safety practices.

Environmental Management System EMS under ISO 14001

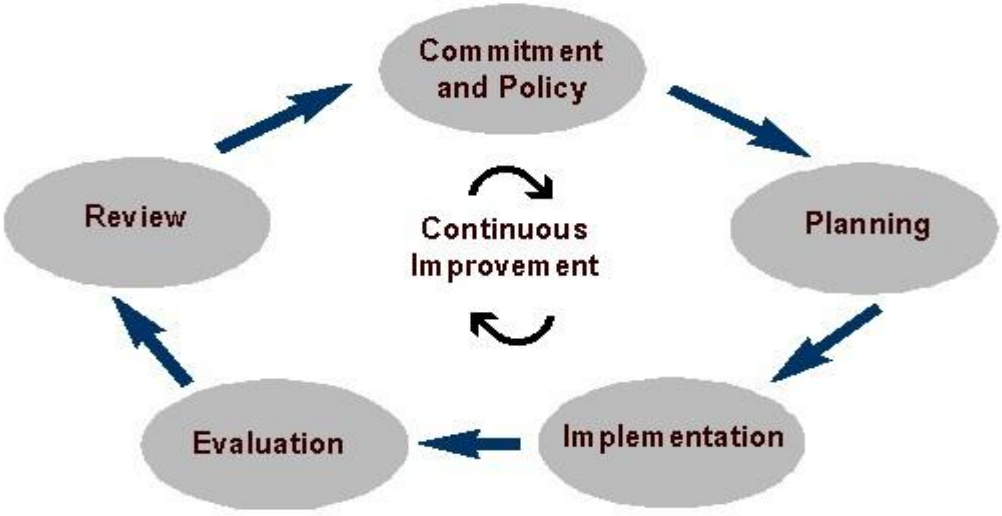


Figure 1: The continuous improvement cycle.

The five main stages of an EMS, as defined by the ISO 14001 standard, are described below:

1. **Commitment and Policy -** Top management commits to environmental improvement and establishes the organization's environmental policy.

2. Planning - An organization first identifies environmental aspects of its operations. Environmental aspects are those items, such as air pollutants or hazardous waste that can have negative impacts on people and/or the environment. An organization then determines which aspects are significant by choosing criteria considered most important by the organization. For example, an organization may choose worker health and safety, environmental compliance, and cost as its criteria.

3. Implementation: A organization follows through with the action plan using the necessary resources (human, financial, etc.). An important component is employee training and awareness for all employees.

4. Evaluation - A company monitors its operations to evaluate whether objectives and targets are being met. If not, the company takes corrective action.

5. Review - Top management reviews the results of the evaluation to see if the EMS is working. Management determines whether the original environmental policy is consistent with the organization's values. The plan is then revised to optimize the effectiveness of the EMS.

- ✓ **Environmental auditing:** Environmental auditing is a systematic, documented, periodic and objective process in assessing an organization's activities and services in relation to: Assessing compliance with relevant statutory and internal requirements. Facilitating management control of environmental practices. the Organic Law No. 04/2005 provides the modalities for the protection, conservation, and promotion of the environment in Rwanda. The requirement for all projects to be subjected to Environmental Impact Assessment (EIA) is stated by the Organic Law in article 67. An environmental audit is a management tool for evaluation of how well Environmental Management Systems (EMS) is performing with the aim of preventing environmental damage; assessing compliance with regulatory requirements; facilitating control of environmental practices by a company/enterprise or facility management; and placing environmental information in the public domain.
- ✓ **Environmental labeling:** Environmental labels provide information about a product or service in terms of its overall environmental benefits, such as the recyclability of its packaging, or the absence of noxious ingredients. Environmental labels provide information about a product or service in terms of its overall environmental benefits, such as the recyclability of its packaging,

or the absence of noxious ingredients. At a time when more and more consumers are motivated to purchase environmentally benign products, this information can positively influence consumer choices, creating a win-win situation for all. However, the increasing proliferation of such labeling has led to concerns about green washing' and exaggerated marketing claims. What is needed, therefore, are internationally agreed and harmonized criteria and methods of labeling to provide a credible and level playing field. This is where ISO International Standards play a fundamental role.

Type I environmental labeling ▸ *for eco-labeling schemes where there are*

clearly defined criteria for products

· **Type II self-declared environmental claims** ▸ *for products and services where*

there are neither criteria nor labeling schemes · **Type III environmental**

declarations ▸ *for specific aspects of products using a life-cycle approach*

- ✓ **Life cycle assessment**: E-LCA is a time-tested assessment technique that evaluates environmental performance throughout the life cycle of a product or from performing a service. The extraction and consumption of resources (including energy), as well as releases to air, water, and soil, are quantified throughout all stages. Their potential contribution to environmental impact categories is then assessed. These categories include climate change, human and eco-toxicity, ionizing radiation, and resource base deterioration (e.g. water, nonrenewable primary energy resources, land, etc.).
- ✓ **Environmental indicators:** A parameter or a value derived from parameters that describe the state of the environment and its impact on human beings, ecosystems, and materials, the pressures on the environment, the driving forces, and the responses steering that system. An indicator has gone through a selection and/or aggregation process to enable it to steer action.

List of indicators

- **Air and Climate**

Greenhouse gas emissions

Emissions of air pollutants

- **Biodiversity**

Threatened species

- **Forest**

Depletion and growth of forest resources in terms of volume

- **Waste**

Municipal waste generation and treatment

Generation of the primary waste sector

- **Water**

Freshwater abstractions

Freshwater resources

Wastewater treatment

Lake and river quality

- **Policy instruments**

Environmental protection expenditure and revenues

Instruments used for environmental policy

- **Resources**

Material resources

- **Green growth**

OECD green growth indicators

Environmental policies: Environmental policy is primarily concerned with how to govern the relationship between humans and the natural environment in a mutually beneficial manner. Traditionally, it has been defined in terms of the problems it addressed, such as controlling pollution and waste flows and limiting habitat loss.

This National Environment and Climate Change Policy is a cross-sectoral policy document. The policy goal is for "Rwanda to have a clean and healthy environment resilient to climate variability and change that supports a high quality of life for its society." The seven objectives of the policy are (1) Greening economic transformation (2) Enhancing functional natural ecosystems and managing biosafety, (3) Strengthening meteorological and early warning services (4) Promoting climate change adaptation, mitigation, and response (5) Improving environmental well-being for Rwandans (6) Strengthening environment and climate change governance (7) Promoting green foreign and domestic direct investment and other capital inflows. To implement these policies, 22 policy statements and 127 policy actions have been identified.

- ✓ **Eco-balances:** Ecology is a field of science that specializes in the study of environmental systems and the organisms that live within those systems, collectively called ecosystems. Ecological balance or ecological equilibrium is an ecological concept that describes how ecosystems exist in a dynamic state of balance or equilibrium. Dynamic equilibrium means that despite disturbances, which may be natural or anthropogenic (human-caused), a balanced ecosystem remains stable because it is in a constant state of flux, perpetually compensating for changes. Simply put, a balanced ecosystem is in a state of dynamic stability where different species interact with each other and their environment in a sustainable way.
- ✓ **Environmental reporting:** Environmental reporting is much more typically associated with the reporting of quantitative and detailed environmental data within the non-financial sections of the annual report or in stand-alone (including website-based) Environmental Reports. Such reports might include pollution emissions to land, air, or water, resources used, or wildlife habitat damaged or re-established.
- ✓ **Environmental charters:** Environmental charters provide a guide for responsible environmental decision-making within a company. An environmental charter contains a set of principles relating to the company's environmental management system. By signing a charter, a company publicly declares its intention to carry out its environmental management activities in accordance with these principles.

The general charter requirements are that companies should have**: *environmental management policies, promote environmental technology development, technology transfers, emergency measures, public relations, education, community relations, overseas operations, contribution to public policies, and responses to global problems.***

LEARNING OUTCOME1.2: DESCRIBE THE ENVIRONMENT OF THE EARTH

Earth environment: Earth's environments range from polar ice caps to hot, dry deserts, from dark ocean depths to high mountaintops. The planet's ecosystems involve complex interactions between the physical (non-living) and biological (living) worlds. NSF supports research that leads to an understanding of Earth's environment.

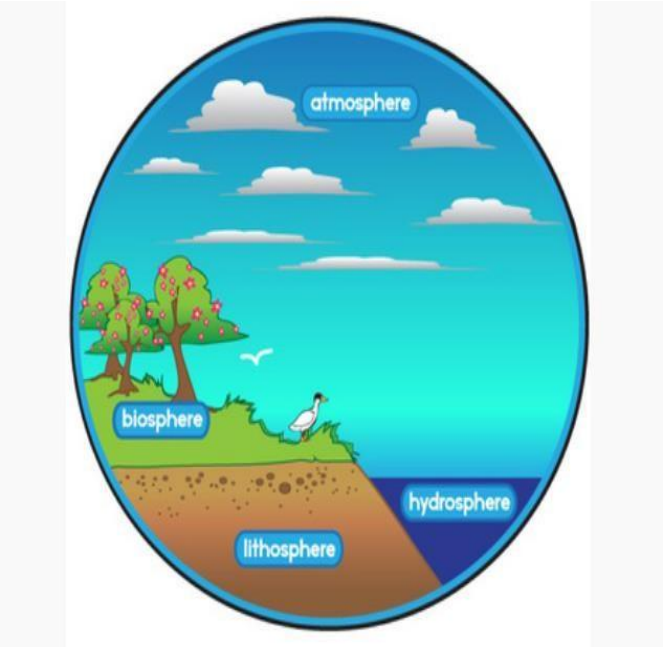
1.2.1. DIFFERENT KEY TERMS

- **Environment:** Environment can be defined as a sum total of all the living and non-living elements and their effects that influence human life. While all living or biotic elements are animals, plants, forests, fisheries, and birds, non-living or abiotic elements include water, land, sunlight, rocks, and air.
- **Environment of Earth:** Everything in our immediate surroundings, including both living and nonliving things like soil, water, creatures, and plants that adapt to their conditions, is referred to as the "environment." It is a gift from nature that helps to sustain life on Earth.

1.2.2. COMPONENTS OF ENVIRONMENT ON EARTH

The three main elements of the environment are the **lithosphere, atmosphere, and hydrosphere**. The lithosphere is the term for the solid portion of the earth. The earth is surrounded by gaseous layers that make up the atmosphere. The atmosphere is composed of gases including oxygen, nitrogen, carbon dioxide, and others. The vast volume of water that makes up the majority of the earth's surface is known as the hydrosphere. This water is present as liquid, ice, and water vapor.

The following image is showing all three elements of environment on Earth



- ✓ **Lithosphere:** The crust, the earth's topmost layer, is composed of several minerals. It can be found on both land (terrestrial crust) and the oceans, where its depth can reach up to 100 kilometers (oceanic crust). Tectonic plates on Earth are the primary part of the lithosphere.
- ✓ **Hydrosphere:** It includes all types of water bodies found on Earth, such as oceans, seas, rivers, lakes, ponds, and streams, among others. On Earth, it covers 70% of the surface. The oceans contain salt water, which makes up 97.5% of the total amount of water on Earth. Freshwater makes up only 2.5% of the world's water. Out of this, 68.9% is frozen in glaciers and 30.8% is accessible as groundwater. Rivers, reservoirs, and lakes have a quantity of 0.3% that is readily available to humans.
- ✓ **Atmosphere:** It is a gaseous layer that surrounds the planet. The amount of oxygen in Earth's atmosphere is unique and necessary for life. In addition to traces of hydrogen, helium, and noble gases, it is mostly composed of 78.08% nitrogen, 20.95% oxygen, 0.93% argon, and 0.038% carbon dioxide. Variable amounts of water vapour are present.
- ✓ **Biosphere:** It describes all areas of the planet where life is present. Ecosystems that support life may exist in the soil, the air, the water, or the land. Geologist Edward Suess came up with the word "biosphere" to describe the area of the planet where life can be found. A biosphere is the whole of all living things, often known as biomass or biota. It stretches from the arctic ice caps to the equator, with some type of life in each area that is appropriate for the environment.

I. Atmosphere (Air)

Definition: The atmosphere is a mixture of gases that surrounds the Earth. It helps make life possible by providing us with air to breathe, shielding us from harmful ultraviolet (UV) radiation coming from the Sun, trapping heat to warm the planet, and preventing extreme temperature differences between day and night. Each of the planets in our solar system have an atmosphere, but none of them have the same ratio of gases or layered structure as Earth's atmosphere.

Atmosphere Composition:

The atmosphere is described as the air that surrounds the earth.

1. The thickness of the earth's atmosphere is about 480 km. 99 percent of the thickness lies up to the height of 32 km from the earth.
2. With increasing altitude, the air pressure decreases.
3. The atmosphere has a mixture of gases that sustains life on earth.
4. The earth's gravity helps hold the atmosphere in place.
5. The major role of the atmosphere is to contain the entry of ultraviolet rays.

The composition of the earth's atmosphere is as mentioned below:

1. Nitrogen — 78 percent
2. Oxygen — 21 percent
3. Argon — 0.93 percent
4. Carbon dioxide — 0.04 percent
5. Trace amounts of neon, helium, methane, krypton, and hydrogen, as well as water vapour

Structure:

The atmosphere is divided into five different layers depending upon the temperature conditions – troposphere, stratosphere, mesosphere, thermosphere and exosphere.

Troposphere

- It is the lowermost layer of the atmosphere.
- The average height of the troposphere is 13 km; its height is about 8 km near the poles and about 18 km at the equator. At the equator, its thickness is greatest because heat is transported to great heights by strong convection currents. **II. Stratosphere**
- It is the second layer of the atmosphere, just above the troposphere and extends up to a height of 50 km.
- This layer of the atmosphere contains the ozone layer which absorbs ultraviolet radiation from the sun and protects life from harmful forms of energy. The UV radiations absorbed by the ozone layer gets converted into heat that is why the stratosphere gets warmer with increasing altitude (unlike the troposphere).

III. Mesosphere

- Mesosphere is the third layer of the atmosphere which extends up to a height of 80 km.
- In this layer, temperature decreases with increasing altitude and drops down to minus 100℃ at the height of 80 km.

IV. Thermosphere

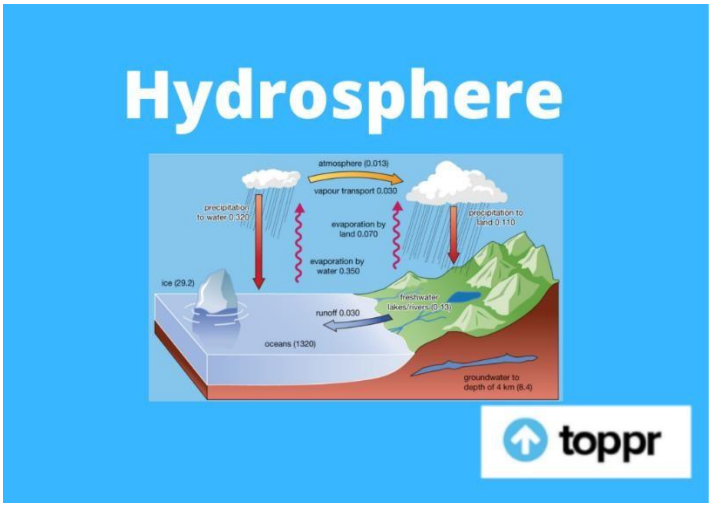
- The ionosphere lies within the thermosphere. It is located between 80 and 400 km above the mesosphere and contains electrically charged particles called ions, hence the name ionosphere.
- In this layer of the atmosphere, temperature increases with increasing height.
- Radio Waves transmitted from the earth are reflected back to the earth by this layer.
- Satellites orbit in the upper part of the thermosphere.

V. Exosphere

- The uppermost layer of the atmosphere above the thermosphere is called the exosphere.
- This layer gradually merges with outer space.

1.2.3. THE HYDROSPHERE (WATER)

Definition: A hydrosphere is **the total amount of water on a planet**. The hydrosphere includes water on the planet's surface, underground, and in the air. A planet's hydrosphere can be liquid, vapor, or ice. On Earth, liquid water exists on the surface in the form of oceans, lakes and rivers.



Composition:

Any water storage area on the Earth that holds liquid water, we consider it to be a part of the hydrosphere. Because of this, there is an extensive list of formations that make up the hydrosphere including:

Oceans: Most of the water on our planet is salt water, and the vast majority of this saltwater is present in the oceans.

Freshwater: Freshwater is much less abundant than saltwater, and is present in a variety of different places.

Surface water: Surface sources of freshwater consist of lakes, rivers, and streams.

Groundwater: Freshwater stored beneath the ground makes up a small portion of the fresh water on Earth.

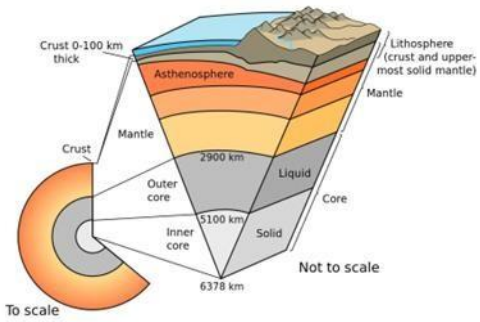
Glacial water: Water that melts off of glaciers.

1.2.4. LITHOSPHERE (LAND)

Definition: The lithosphere is **the rocky outer part of Earth**. It is made up of the brittle crust and the top part of the upper mantle. The lithosphere is the coolest and most rigid part of Earth

Composition

Structure



The lithosphere is the solid, outer part of Earth. The lithosphere includes the brittle upper portion of the mantle and the crust, the outermost layers of Earth's structure. It is bounded by the atmosphere above and the asthenosphere (another part of the upper mantle) below. There are two types of lithospheres: oceanic lithosphere and continental lithosphere. The oceanic lithosphere is associated with the oceanic crust and is slightly denser than the continental lithosphere. Although the rocks of the lithosphere are still considered elastic, they are not viscous.

Advantages

The lithosphere contains different types of rocks such as igneous, sedimentary, and metamorphic rocks, it helps to provide the necessary nutrients required by plants. The mantle, which lies between the core and the crust, consists of molten rock, called magma which moves continuously because of convection currents.

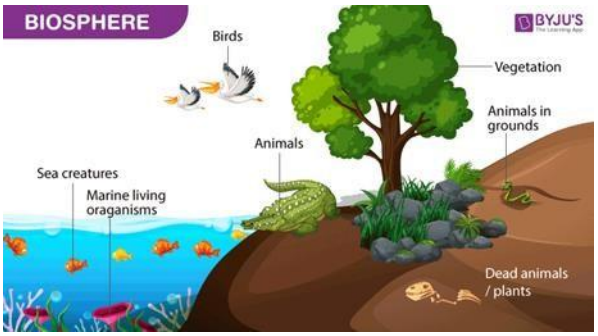
1.2.5. BIOSPHERE (LIVING THINGS)

Definition: The biosphere is **made up of the parts of Earth where life exists**. The biosphere extends from the deepest root systems of trees to the dark environment of ocean trenches, to lush rainforests and high mountaintops.

Composition: The biosphere is made up of the parts of Earth where life exists—all ecosystems. The biosphere extends from the deepest root systems of trees to the dark environments of ocean trenches, to lush rain forests, high mountaintops, and transition zones like this one, where ocean and terrestrial ecosystems meet. ["Spheres" refers to these four subsystems. The **"lithosphere" (land), "hydrosphere" (water), "biosphere" (living things), and "atmosphere" (air)** are the four spheres (air)]

Structure

The biosphere, which includes the ground and the air, is characterized as the region of the planet where organisms live. The biosphere is defined as the region on, above, and below the Earth's surface where life exists.



The biosphere is characterized as an area that contains all living organisms and the products of their activities. As a result, it plays a critical role in the maintenance of ecosystems, i.e., the existence of species and their reciprocal interactions. And the biosphere is critical for climate regulation.

Advantages: It helps to improve a region environmentally, economically and socially but it does have its adverse effects.

LEARNING OUTCOME1.3: DESCRIBE RWANDA'S ENVIRONMENTAL FEATURES

1.3.1. NATURAL ENVIRONMENT

1.3.1.1. GEOGRAPHICAL FEATURES

- **Relief:** Rwanda has hilly and mountainous relief with an altitude ranging between 900 m and 4.507 m, and it has a tropical temperate climate due to its high altitude.
- **Climate:** Rwanda enjoys a tropical temperate climate due to its high altitude. The average annual temperature ranges between 16°C and 20°C, without significant variations. Rainfalls are abundant although they present some irregularities. Winds are generally around 1-3 m/s.
- **Hydrography:** Rwanda possesses a relatively big quantity of water (surface and underground water): rivers, lakes, and marshlands occupy a surface of 211000 ha, which is about 8% of the national territory (lakes: 128000 ha, rivers: 7260 ha and marshlands: 77000 ha). The outflow of the renewable underground resource is estimated at 66 m^3/s, out of which 9 m^3/s are produced by 22000 known sources. In general, too little information is available, especially on underground resources.
- **Vegetation:** Natural vegetation refers to the plants that grow in a given area without human influence. This type of vegetation is not planted. They grow on their own. The specific examples of natural vegetation in Rwanda include; Nyungwe Forest, Gishwati Forest, Mukura Forest, Birunga Bamboo Forest, and moorland and Akagera grasslands. The natural vegetation cover in Rwanda occupies about 15% of the general land area. This leaves the remaining part of the country's land area under other activities such as farming and different socio-economic activities.
- **Natural resources and Soil features**

Rwanda is one of the world's largest producers of tin, tantalum, and tungsten (3Ts) and also exports gold and gemstones. Rwanda also possesses a variety of minerals such as silica sands, kaolin, vermiculite, diatomite, clays, limestone, talcum, gypsum, and pozzolan. They form the main type of soil in Rwanda. These soils developed on weathered parent material which mainly had clay mixed with iron and aluminum oxides. They appear reddish due to the presence of iron and aluminum oxides. The profile of kaolisols reveals an upper humus horizon of low humus while the

structure of the middle section differs for different subtypes, but it generally shifts gradually from the humus horizon to the parent rock. Kaolisols are found in the central area of Rwanda. They are usually deep.

- **Wetlands**

Wetlands (Kinyarwanda terms, “Ibishanga n’ibibaya” or “ubutaka buhehereye”) play a vital role in maintaining water resources and are part of the green infrastructure of healthy catchments. They protect water quality, maintain water supplies by replenishing rivers and streams during the dry season, and more importantly, reduce the risk of flooding, sequester carbon, and play many critical roles in Rwandan ecosystems. In addition, Rwandan wetlands offer a habitat for a vast array of biological diversity that cannot survive elsewhere.

- **Forests and protected areas**

Rwanda’s forests and woodlands can be classified into four categories: the natural forests of the Congo Nile Ridge comprised of Nyungwe national park Gishwati, and Mukura; the natural forests of the Volcanoes national park; the natural forests in the savannah and gallery-forest of the Akagera national park and remnants of gallery forests and savannahs of Bugesera, Gisaka and Umutara; and forest plantations dominated by exotic species (Eucalyptus sp, Pinus sp, Grevillea robusta) and trees scattered on farmlands (agroforestry) and along antierosion ditches.

Protected areas

According to the Convention on Biological Diversity, a protected area is defined as a portion of land, aquatic or sea environment which is geographically delimited, dedicated especially to the protection and the preservation of biological diversity and its natural and cultural resources; hence this geographic area must be legally indicated, regulated and be managed by effective, legal means or others***. According to this definition of protected areas, Rwanda has four types of protected areas which include national parks (Akagera, Nyungwe, and Volcanoes National Park); forest reserves (Gishwati, Iwawa Island and Mukura forest reserves); forests of cultural importance (Buhanga forest); and wetlands of global importance (Rugezi-Bulera-Ruhondo wetland complex). Besides those forests with a legal status of protected areas, there are other forests of cultural importance (Busaga forest in Muhanga district) and other remnant natural forests which are more or less protected by law.***

- **Human environment**

Major current environmental issues may include climate change, pollution, environmental degradation, and resource depletion. The conservation movement lobbies for the protection of endangered species and protection of any ecologically valuable natural areas, genetically modified foods, and global warming.

- **Socio-demographic and economic features**

Economic aspect

Human settlement

LEARNING UNIT 2–ASSESS ENVIRONMENTAL IMPACTS

Learning Outcomes:

2.1 Determine the environmental impact

2.2 Conduct Environmental Impact Assessment

2.3 Develop an Environmental Impact Assessment (EIA) report

LEARNING OUTCOME 2.1: DETERMINE THE ENVIRONMENTAL IMPACT

2.1.1. Environmental aspects

The Standard defines an environmental aspect as 'An element of an organization's activities or products or services that interacts or can interact with the environment. These can include **discharges to water, emissions to air, waste, and the use of natural resources and materials**. For example, one of the environmental aspects of **car washing may be a cleaning agent that has the potential for water pollution** (this pollution is the environmental impact).

Environmental Impact

An environmental Impact (EI) is defined as any change to the environment, whether adverse or beneficial, resulting from a facility's activities, products, or services. In other words, it is the effect that people's actions have on the environment. Humans impact the physical environment in many ways: **overpopulation, pollution, burning fossil fuels, and deforestation**. Changes like these have triggered climate change, soil erosion, poor air quality, and undrinkable water.

There are three main impacts on the environment: **global warming and climate change; water pollution and ocean acidification; and loss of biodiversity.**

2.1.2. Categories of environmental impacts

Unwanted Emissions to the air

Emissions are a type of air pollution. They are **any kind of substance released into the air from natural or human sources, flows of gases, liquid droplets or solid particles**. Not all emissions become air pollutants, but many do, causing significant health and environmental problems. The criteria pollutants are **carbon monoxide (CO), lead (Pb), nitrogen dioxide (NO_2), ozone (O_3), particulate matter (PM), and sulfur dioxide (SO_2)**. There are also a large number of compounds that have been determined to be hazardous which are called air toxics. These unwanted emissions in the air results into global warming.

Global warming: is the long-term heating of Earth's surface observed since the pre-industrial period (between 1850 and 1900) due to human activities, primarily fossil fuel burning (**the burning of oil, natural gas, and coal to generate energy**), deforestation, farming, and industrial emissions, which increases heat-trapping greenhouse gas levels in Earth's atmosphere.

Pollution of land

Land pollution refers to **the destruction or a decline in the quality of the earth's surface as a result of human actions**. This could include anything from mining to agriculture to building, and consequences may be intentional or unintentional. Land pollution includes legacy pollution, illegal dumping, and litter.

Discharge and pollution of water

The discharge is **the volume of water that is transported each second**, and the volumes are expressed in liters (l) or in cubic meters (m3) [1 m3 = 1000 l], and the letter Q is commonly used as the symbol for the discharge.

Depletion of materials/natural resources

We eat edible plants in the form of leaves, fruit, and vegetables, some of these resources are extinct by humans.

We use glass for windows, television screens, drinking glasses, eyeglasses, and car windshields.

We use metals for car bodies, tin cans, appliances, utensils, faucets, piping, and in electrical wires.

Solid waste management

The term solid waste management mainly refers to **the complete process of collecting, treating and disposing of solid wastes**. In the waste management process, the wastes are collected from different sources and disposed of. This process includes collection, transportation, treatment, analysis and disposal of waste.

Hazardous waste

Hazardous waste is **waste with properties that make it dangerous or capable of having a harmful effect on human health or the environment**." Hazardous waste also includes different physical forms, including solids, liquids, and gases.

In order to keep as much material out of the landfill as possible, it's important for each of us to do our part. One of the ways to put that plan into action is through the 3 Rs of waste management

— **Reduce, Reuse, Recycle**

✓ **Categories of environmental impact**

Global warming

Global warming is an aspect of climate change, referring to the long-term rise of the planet's temperatures. It is caused by **increased concentrations of greenhouse gases in the atmosphere, mainly from human activities such as burning fossil fuels, and farming**.

5 effects of global warming: **More frequent and intense drought, storms, heat waves, rising sea levels, melting glaciers, and warming oceans** can directly harm animals, destroy the places they live, and wreak havoc on people's livelihoods and communities.

Ozone depletion

Ozone Depletion. **When chlorine and bromine atoms come into contact with ozone in the stratosphere, they destroy ozone molecules**. One chlorine atom can destroy over 100,000 ozone molecules before it is removed from the stratosphere. Ozone can be destroyed more quickly than it is naturally created.

Ozone depletion **can cause increased amounts of UV radiation to reach the Earth which can lead to more cases of skin cancer, cataracts, and impaired immune systems**. Too much exposure to UV is believed to be contributing to the increase in melanoma, the most fatal of all skin cancers.

Climate change is concerned with how carbon dioxide, methane, and other greenhouse gases are altering the global climate system. Ozone depletion, on the other hand, **involves how certain industrially produced chemicals containing chlorine or bromine are damaging the earth's protective stratospheric ozone layer.** **Acidification of soil and water**

Soil acidification is **a process where the soil pH decreases over time**. This process is accelerated by agricultural production and can affect both the surface soil and subsoil. Queensland has more than 500,000 hectares of agricultural and pastoral land that is acidified or is at risk of acidification. There are four major reasons for soils to become acidic: **rainfall and leaching, acidic parent material, organic matter decay, and harvest of high-yielding crops**. Wet climates have a greater potential for acidic soils. **Eutrophication**

Eutrophication is **a natural process that results from the accumulation of nutrients in lakes or other bodies of water**. Algae that feed on nutrients grow into unsightly scum on the water surface, decreasing recreational value and clogging water-intake pipes. eutrophication, the gradual increase in the concentration of phosphorus, nitrogen, and other plant nutrients in an aging aquatic ecosystem such as a lake. The productivity or fertility of such an ecosystem naturally increases as the amount of organic material that can be broken down into nutrients increases. **Photochemical smog**

Photochemical smog is **a mixture of pollutants that are formed when nitrogen oxides and volatile organic compounds (VOCs) react to sunlight, creating a brown haze above cities**. It tends to occur more often in summer because that is when we have the most sunlight. Primary pollutants. Photochemical smog is produced when **sunlight reacts with nitrogen oxides and at least one volatile organic compound (VOC) in the atmosphere**. Nitrogen oxides come from car exhaust, coal power plants, and factory emissions. VOCs are released from gasoline, paints, and many cleaning solvents.

Depletion of abiotic resources-elements

Abiotic depletion refers to **the depletion of nonliving (abiotic) resources such as fossil fuels, minerals, clay, and peat**. Abiotic depletion is measured in kilograms of Antimony (Sb) equivalents. Resources of abiotic factors are usually obtained from the atmosphere, lithosphere, and hydrosphere. Examples of abiotic resources are **air, water, sunlight, soil, and minerals**.

Depletion of abiotic resources-fossil fuels

Abiotic depletion refers to the depletion of nonliving (abiotic) resources such as fossil fuels, minerals, clay, and peat. Abiotic depletion is measured in kilograms of Antimony (Sb) equivalents. Global warming potential is an appraisal of greenhouse gas (for example, CO_2, methane, nitrous oxide…) Fossil fuel depletion is **the reduction in future availability of fossil fuels caused by the primary extraction of fossil fuels linked to fuel use, energy use and to produce other inputs, such as mineral fertilizer**. Fossil fuels are a special case in this regard. For while they are derived from biomass that was produced millions of years ago, by virtue of their having attained their current state through extremely gradual geological processes, they qualify as abiotic resources.

- ✓ **Human toxicity**

The human toxicity potential (HTP), **a calculated index that reflects the potential harm of a unit of chemical released into the environment**, is based on both the inherent toxicity of a compound and its potential dose. The toxicity of a substance depends on three factors: its chemical structure, the extent to which the substance is absorbed by the body, and the body's ability to detoxify the substance (change it into a less toxic substance) and eliminate it from the body.

- ✓ **Freshwater aquatic ecotoxicity**

The Freshwater Aquatic Ecotoxicity Potential (FAETP) is **the toxic effects of chemicals on an ecosystem, in this case in the freshwater, causing biodiversity loss and/or species extinction**.
The characterization factors are expressed like ETP in kg1

- ✓ **Marine aquatic ecotoxicity**

Ecotoxicology is **the study of toxic effects caused by natural or man-made substances on biota**. Through ecotoxicology, we are able to determine the levels and types of contaminants that cause harm to animals and plants.

Marine ecotoxicity, as per definition, is about the **toxic substances' impact on the marine ecosystem**, and it originates from air emission of electric supply and non-ferrous metal

- ✓ **Terrestrial ecotoxicity**

Terrestrial ecotoxicity refers to the environmental pollutants that affect land-dependent organisms and their environment. It requires three elements: (1) a source, (2) a receptor, and (3) an exposure pathway. This article reviews the basic principles of each element as they occur in terrestrial

systems. Pollutants enter the terrestrial environment through direct application, diffuse sources, or long-range transport. Terrestrial receptors include soil microbes, invertebrates, plants, amphibians, Rediff sources tiles, birds, and mammals. Terrestrial organisms can be exposed to pollutants through dermal, oral, inhalation, and food-chain exposures. Two important processes govern the movement of a pollutant into and through a terrestrial food web: bioaccumulation and biomagnification.

- ✓ **Water pollution**

Water pollution is the contamination of water sources by substances that make the water unusable for drinking, cooking, cleaning, swimming, and other activities. Pollutants include chemicals, trash, bacteria, and parasites. All forms of pollution eventually make their way to the water. Water pollutants may cause disease or act as poisons. Bacteria and parasites in poorly treated sewage may enter drinking water supplies and cause digestive problems such as cholera and diarrhea. Hazardous chemicals, pesticides, and herbicides from industries, farms, homes, and golf courses can cause acute toxicity and immediate death, or chronic toxicity that can lead to neurological problems or cancers.

- ✓ **Air pollution**

Air pollution is the contamination of the indoor or outdoor environment by any chemical, physical or biological agent that modifies the natural characteristics of the atmosphere.

Household combustion devices, motor vehicles, industrial facilities and forest fires are common sources of air pollution. Pollutants of major public health concern include particulate matter, carbon monoxide, ozone, nitrogen dioxide, and sulfur dioxide. Outdoor and indoor air pollution cause respiratory and other diseases and are important sources of morbidity and mortality.

2.1.3. Procedures for determining environmental aspects and impacts

Identify the process associated with the activities of the workplace

Once you understand compliance obligations, you should assess how your organization interacts with the environment. Identify all of your environmental aspects and impacts, and determine which are significant. Some environmental aspects may be regulated, while others may not be. For example, if you identify the generation of a particular air emission as a significant environmental aspect, it would help to know which operation(s) generate such air emissions.

It might also help to know whether these air emissions are monitored or otherwise measured in some manner. **Consider the potential and actual impacts on the environment from each process and subprocesses**

Humans impact the physical environment in many ways: overpopulation, pollution, burning fossil fuels, and deforestation. Changes like these have triggered climate change, soil erosion, poor air quality, and undrinkable water. These negative impacts can affect human behavior and can prompt mass migrations or battles over clean water.

Identify the environmental aspects as the source of each impact to the environment

Identify legal requirements

Identify the potential for emergency situations

Consider the environmental aspects associated with legal requirement

Consider the ease of changing the environmental impacts

LEARNING OUTCOME2.2: CONDUCT ENVIRONMENTAL IMPACT ASSESSMENT

2.2.1. Environmental Impact Assessment (EIA)

According to **United Nations Environment Programme (UNEP**) defines **Environmental Impact Assessment (EIA)** as a tool used to identify the environmental, social, and economic impacts of a project prior to decision-making. It aims to predict environmental impacts at an early stage in project planning and design, find ways and means to reduce adverse impacts, shape projects to suit the local environment, and present the predictions and options to decision-makers.

According to the Rwanda Environmental law **N°48/2018 of 13/08/2018** determines modalities for protecting, conserving and promoting the environment, in article two.

As Rwanda a country which considers the lives of the population as the primarily resources and according to the Article 30 of Rwanda Environmental Law No. 48/2018, the Ministerial Order No 001/ 2019 of 15/04/2019 and the World Bank Environmental and Social Standards on Assessment and Management of Environmental and Social Risks (ESS 1). In different projects, especially some supported by World Bank, they make **Environmental and Social Impact Assessment (ESIA).**

Environmental and Social Impact Assessment (ESIA) is defined as predicts of the environmental and social consequences that a future development project/intervention might entail. It is carried out before project implementation and proposes measures to mitigate potential negative impacts.

Environmental impact assessment: a systematic process of identifying environmental, social and economic impacts of a project before a decision of its acceptance is made.

According to this law also, in its article number 30, they list all development projects **that must undergo an environmental impact assessment and its procedure.**

The list of projects that must undergo an environmental impact assessment before they obtain authorization for their implementation is established by an Order of the Minister. An Order of the Minister also issues instructions and procedures for conducting environmental impact assessment. The ministerial order **No 001/ 2019 of 15/04/2019** determines establishing the list of projects that must undergo environmental impact assessment, instructions, requirements and procedures to conduct environmental impact assessment.

2.2.2. LIST OF WORKS, ACTIVITIE AND PROJECTS THAT MUST UNDERGO A FULL ENVIRONMENTAL IMPACT ASSESSMENT

The projects are the following:

1° all buildings classified as residential, commercial, administrative or institutional sports facilities, social, cultural, and assembly and religious buildings, hotels, health facilities, educational buildings, or other publicly accessible facilities fulfilling at least two of the following conditions:

a. having capacity to host more than five hundred (500) people;

b. having a total floor area exceeding one thousand and five hundred square meters (1500 sqm);

c. built in plot size exceeding one thousand square meters (1000 sqm).

2° industries;

3° buildings with activities which can cause hazards;

4° warehouses for storage of hazardous materials and perishables;

5° warehouses and storage facilities of perishable agricultural commodities occupying an area of a half a hectare (1/2 ha) and above;
6° places where people are detained for punitive or corrective reasons or because of their mental condition;
7° places of non-industrial commercial activities, where such activities or commercial products may cause fire, cause combustion with extreme rapidity, give rise to poisonous fumes or cause explosions;
8°garages for vehicle repair and maintenance
9° warehouse with total floor area exceeding one thousand and five hundred square meters (1500 sqm) and plot size exceeding one thousand square meters (1000 sqm);
10° hotels;
11° construction and repair of international, national, District roads and repair of large bridges;
12° construction of hydro-dams, hydropower plants and electrical lines of high and medium voltage;
13° construction of petroleum products and gas pipelines and their storage tanks;
14° construction of terminal ports, airports and railways;
15° agricultural and breeding activities which use chemical fertilizers and pesticides in wetlands;
16° large scale investment projects in agricultural and breeding activities which use chemical fertilizers and pesticides on hill sides;
17° works and activities that involve genetic modification of seeds and animals
18° works in areas designated as national parks and in their buffer zones;
19° all works related to mining;
20° commercial small and large scale quarrying;
21° construction of artificial lakes;
22° construction of water storage dams;
23° works or activities for marshlands reclamation; 24°
construction of slaughter houses.

Purpose for making EIA: An impact assessment is a planning and decision-making tool used **to assess the potential positive and negative effects of proposed projects**. Impact assessments consider a wide range of factors and propose measures to mitigate projects' adverse effects **Importance:** EIAs make sure that project decision-makers think about the likely effects on the

environment at the earliest possible time and aim to avoid, reduce or offset those effects. This ensures that proposals are understood properly before decisions are made.

The major environmental issues surrounding our environment:

Climate Change and global warming

- **Pollution of air,**
- **Deforestation,**
- **Industrialization**
- **Overfilling landfills**
- **Greenhouse gases**

Water pollution

Improper sewage disposal

- **Oil spills**
- **Disposal of chemical**
- **Radioactive waste**
- **Plastic pollution**

Loss of biodiversity and Land Use

Biological resources: Biodiversity is all the different kinds of life you'll find in one area—the variety of animals, plants, fungi, and even microorganisms like bacteria that make up our natural world. Each of these species and organisms works together in ecosystems, like an intricate web, to maintain balance and support life.

Types of EIA Principles

Polluter Pays Principle (PPP): ensuring absolute liability for any damage and harm caused by the industry and firms. It makes the process and procedure for compensation easy in an event where their victims are affected. Another important aspect of this among principles of environmental management is that the cost gets shared and it's easy to repair or reduce damages.

The Precautionary Principle (PP): The Precautionary Principle. The precautionary principle states that if a product, an action, or a policy has a suspected risk of causing harm to the public or to the environment, protective action should be supported before there is complete scientific proof of a risk.

2.2.3. SOME USED TERMS IN ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

Authority: Is a national body responsible for commissioning, supervising and approving the EIA report and issuing the EIA Certificate to the Developer, Here, the authority is Rwanda Development Board (RDB) in partnership with Rwanda Environmental Management Authority (REMA).

Environmental impact study: Is an evaluation report of the impact of a planned activity on the environment;

Developer: Is a person, group of persons or agency developing a new development project which is subject to an Environmental Impact Assessment process.

Expert: A person with special knowledge or ability who performs skillfully. For EIA in Rwanda, experts are published by REMA.

Lead Agency: It is any public office or organization including every Ministry or government department which has functions or responsibilities for the protection of any segment of the environment, conservation and sustainable use of natural resources.

License: It is a permit that is given to the developer for performing different development activities in different sectors.

Project Brief: It is a summary statement of the likely environmental effects of a proposed development and it includes description of the site and proposed development in sufficient detail to enable the Authority to determine whether an EIA is required or not.

2.2.4. STEPS OF THE EIA PROCESS

Project proposal: The proposed project consists of a written document outlining everything stakeholders should know about a project, including the timeline, budget, objectives, and goals. Your project proposal should summarize your project details and sell your idea so stakeholders buy in to the initiative.

Screening: Screening is the first stage of the EIA process which results in a key EIA decision, namely to either conduct the assessment (based on the likely significant impacts) or not conduct it (in the anticipated absence of such impacts).

Screening is also defined as a **process by which a decision is taken on whether or not EIA is required for a particular Project**.

The EIA Procedure Guidelines categorises projects into three categories or sometimes referred to as levels of impact:

- **Category 1 (Impact level 1):** Full EIA not required. IL-1 projects are not subjected to Environmental Impact Study, instead they go directly to the Decision-making and Authorization stage. IL-1 projects are however subjected to a period of public review during which stakeholders may submit written views to the Authority.
- **Category 2 (Impact level 2):** Mitigation measures for IL-2 projects will be based on nature of the project, its components and input of the review committees
- **Category level 3 (Impact Level 3):** projects need to pass through the full Environmental Impact Study process.

Screening needs to follow specific procedures often described in the legislation so all the projects follow the same process.

Key contributions of screening to a good EIA:

- Facilitates informed decision-making by providing clear, well-structured, factual analysis of the effects and consequences of proposed actions.
- Influences both project selection and policy design by screening out environmentally and/or socially unsound proposals, as well as modifying feasible action.

Scoping: *Scoping* is a critical step in the preparation of an EIA, as it identifies the issues that are likely to be of most importance during the EIA and eliminates those that are of little concern. Scoping is a systematic exercise that establishes the boundaries of your EIA and sets the basis of the analyses you will conduct at each stage. A quality scoping study reduces the risk of including inappropriate components or excluding components that should be addressed. It involves:

- Identifying all relevant issues and factors, including cumulative effects, social impacts, and health risks.
- Facilitating meaningful public engagement and review.
- Determining the appropriate time and space boundaries of the EIA.
- Identifying the important issues to be considered in the EIA, such as setting the baseline and identifying alternatives.

EIA process involves the following four stages which are:

1. **The Environmental Impact Initiation phase**, which involves: screening and scoping;
2. **The Impact Study Phase**, which includes: impact identification and analysis, development of mitigation measures and preparation of the report;
3. **The decision-making & authorization phase entails review of EIA reports leading to decision-making**;

Environmental management Plan and follow-up phase: deals with monitoring and costing of activities of the project during its implementation

Impact Assessment and Mitigation: Impact assessment refers to the detailed evaluation of the environmental and social impacts of the planned project and identified alternatives, compared to the baseline conditions. This includes qualitative descriptions such as measuring high, medium, and low impacts, and quantitative descriptions, such as indicating the cubic meters of water withdrawn, sewage produced, and pollutants released. This is done for the planned project as well as the identified alternatives, allowing for comparisons. Once the detailed assessment is complete, mitigation measures to reduce or avoid impacts are identified.

Mitigation refers to minimizing or avoiding the described impacts. Overall, mitigation measures are a response to the findings of impact assessment; they need to cover all the areas identified. The key focus of mitigation actions should be on:

- Preventive measures that avoid the occurrence of impacts and thus avoid harm or even produce positive outcomes.
- Measures that focus on limiting the severity and the duration of the impacts.
- Compensation mechanisms for those impacts that are unavoidable and cannot be reduced further.

Key impacts and potential mitigation actions often relate to the land. Almost all development proposals involve disturbance of the land surface. This is usually extensive for major linear projects (roads, pipelines), dams and reservoirs, and large-scale mining, agriculture, forestry, and housing schemes. Environmental impacts of particular concern can include the drainage of wetlands, conversion of natural areas, or expansion into areas that are vulnerable to natural hazards.

Impact management: Essentially, *impact management* is the creation of a series of plans and protocols aiming to manage and monitor the identified mitigation measures and risks that may occur over the project lifetime, such as technical failures and natural disasters. Some of the plans are compulsory, such as an environmental management plan (EMP), which is required as part of an EIA report in most countries. Other plans are context-specific and/or depend on guidance from national legislation. For example, if the project takes place in close proximity to a community, a resettlement plan may be needed; if there are no communities close by, a resettlement plan would not be necessary. Impact Management begins during the project's planning phase and continues on after project implementation.

Environmental impact statement (EIS): An Environmental Impact Statement (EIS) is a document prepared to describe the effects of proposed activities on the environment. "Environment," in this case, is defined as the natural and physical environment and the relationship of people with that environment. This means that the "environment" considered in an EIS includes land, water, air, structures, living organisms, environmental values at the site, and social, cultural, and economic aspects. An "impact" is a change in consequence that results from an activity. Impacts can be positive or negative or both. An EIS describes impacts, as well as ways to "mitigate" impacts. To "mitigate" means to lessen or remove negative impacts.

- ✓ **EIS Reporting:** The EIA Report is a compilation of several important project components, including the project description, the assessment of its environmental and social impacts, mitigation measures, and related management and monitoring plans. During this step, all the information gathered during the previous steps is compiled into a comprehensive report that analyzes and synthesizes the data, structuring it as stipulated in the terms of reference (TOR). A quality EIA Report has the following characteristics:
  - It is well-structured and uses non-technical language supported by data and well-executed analyses.
  - It provides information that is helpful and relevant to decision-making.
  - It results in the satisfactory prediction of the adverse effects of the proposed actions and their mitigation using conventional and customized techniques.

EIS Review & Licensing: Review and Licensing is the final check on the quality of the EIA report submitted to obtain a project license. Once the EIA report is submitted, designated authorities will usually go through it thoroughly, weighing the methods used, data, interpretations, measures, and conclusions to assess the impacts of the planned development. Their review will determine whether or not the project adequately addresses major environmental and social impacts and other risks, and whether or not to grant a license to the project proponents or to request project changes. This means that a good quality EIA might still lead to the planned development not being permitted to go ahead based on the identified impacts. Often, the review process leads to a requirement for additional information on potential impacts, mitigation measures, or other aspects.

Monitoring: Monitoring provides data on the environmental and social impacts of the project for the whole project lifecycle. As part of their operations, most development projects involve regular monitoring of indicators (including such things as the quantity of ores extracted, materials processed, energy used, and sewage released, etc.). More specifically, the information collected during monitoring activities helps to ensure that the priorities listed in the Environmental Management Plan (**EMP**), mitigation measures, and contingency plans are properly implemented and that these plans and measures are effective in addressing the project's impacts.

2.2.5. KEY INSTITUTIONS IN THE EIA FOR DEVELOPMENT PROJECTS

No	KEY INSTITUTION/ AGENCY	ROLES / RESPONSIBILITIES IN THE EIA PROCESSES FOR DEVELOPMENT PROJECTS
1	Ministry for Environment (MINIRENA) and Ministry of Land and Forestry	Formulating policies, laws and standards for land administration and land use planning; environmental protection and natural resources utilization.
2	Land management and use	Mapping, and technical support to and regulation for all lands
3	Rwanda Environmental Management Authority (REMA)	National authority responsible for environmental regulations and standards setting, and overseeing the implementation of EIA guidelines. REMA is responsible for monitoring the implementation of EMPs for approved projects as part of the compliance enforcement; and ensuring that the EIA process is undertaken as per the law and established standards.
4	Rwanda Development Board (RDB)	RDB One Stop Centre issues licenses to investors, commissions and approves EIA in collaboration with REMA. In the EIA process, RDB reviews the EIA reports, organizes site visits and public hearing; and ensures that due diligence is made in the EIA process before the certificate is issued.
5	Rwanda Development Board (RDB)	RDB One Stop Centre issues licenses to investors, commissions and approves EIA in collaboration with REMA. In the EIA process, RDB reviews the EIA reports, organizes site visits and public hearing; and ensures that due diligence is made in the EIA process before the certificate is issued.

6	Ministry of Disaster Management (MIDMAR)	Policy, planning and coordination of disaster prediction and response in different activities/ areas
7	Ministry of Trade and Industry (MINICOM)	Trade and industries policy and laws
8	Local Authorities	Implementation of land use zoning ; local revenue
9	Rwanda Bureau of Standards	Setting standards, setting and certification of quality of products
10	Ministry of Health (MINISANTE)	Responsible for setting policy and guidelines for environmental and public Health/safety and sanitation.
11	Ministry of Public Service and Labour	Setting the labour and employment policies/laws, inspection of service
12	Private Sector Federation	Mobilizing and sensitizing members involved in the business of different types
13	Rwanda Cooperatives Agency (RCA)	Mobilization, registration and capacity building for some Cooperatives and associations
14	Workers' Union (CESTRAR)	Ensuring that the workers' rights and responsibilities are upheld with respect to remuneration and welfare; occupational health and safety
15	EIA Experts	Require tools and information conduct EIA in a professional and independent manner; advise Developers to follow the standards and regulations of EIA
16	Construction firms	These companies are doing different jobs in different construction activities such as plans, housing and road, bridges, industries and other work in construction

LEARNING OUTCOME 2.3: DEVELOP AN EIA REPORT

2.3.1. ENVIRONMENTAL IMPACT STUDY AND REPORTING

The EIS which is a research and investigation phase of the EIA process is the main stage of intervention. **For development projects, it involves a three-step process:**

1. ***Potential impacts of a project and their magnitude are identified*.** Also included in this step is the Analysis of Initial State. IL-3 projects start the EIS process at step 1 while IL-2 projects start the process at Step 2. IL-1 projects are not subjected to EIS, and instead they go directly to the Decision-making and Authorization stage. IL-1 projects are however subjected to a period of public review during which stakeholders may submit written views to the Authority.
2. ***An Environmental Impact Report including an Environment Management Plan (EMP)*** is drafted on completion of the investigations. The main objective of an EMP is to streamline environmental issues into the business and operational plans of the project. An EMP is incorporated into the Environmental Impact Report and submitted to the developer who may, if necessary, append an addendum (*Environmental Impact Report Addendum)* to the EIA report. The developer then submits the EIA report to the Authority, which checks for completeness before passing them on to Lead Agencies and stakeholders for review (Step 3).
3. ***The EIR is subjected to a formal public hearing and post-hearing consultation.*** Output of the public hearing is a Public Hearing Report, written by the presiding officer (RDB staff). The public hearing report, EIR and the developer's response, constitute the basis for decision making regarding approval or disapproval of the project.

2.3.2. SOME KEY TASKS THAT MUST BE PERFORMED BY THE EIA EXPERTS DURING IMPACT STUDY ARE:

1. **Analysis of the initial state:** During environmental impact study, EIA Experts should undertake an analysis of the initial state of the environment performed to create a comparative basis of impacts after project implementation commences. Analysis of Initial

State should include a record of baseline environmental conditions considered to be threatened by the project.

2. **Identification and Analysis of Impacts:** This involves prediction and analysis of potential socio-environmental impacts that would result from developing, operation and decommissioning of the project.
3. **Mitigation Measures, Alternatives and Monitoring:** This entails identification and assigning responsibilities and duties related to impact mitigation, alternative project options and requirements for monitoring. After mitigation measures have been identified, viable alternatives considered, details and schedule for monitoring during project implementation identified, the EIA Experts shall include this information in the Environmental Impact Report.
4. **Mitigation Measures:** Mitigation measures are intended to prevent or minimize negative impacts of a project and enhance the positive ones. EIA Experts shall develop mitigation measures for IL-3 projects, basing on findings of the environmental impact study. Mitigation measures for IL-2 projects will be based on nature of the project, its components and input of the review committees. The EIA experts shall prioritise mitigation measures, organizing them into a hierarchy of importance with highest priority given to measures that prevent highly significant adverse environmental or socioeconomic impacts.
5. **Review of Alternatives:** During EIA studies, the EIA experts shall undertake an analysis of alternatives with the view of finding feasible ways to prevent or minimize negative impacts while maintaining project objectives.

6. **Preparation of EIA Report:** The EIA experts shall compile results of an impact study into a report termed an *Environmental Impact Report (EIR)*.

This document should provide the Authority with sufficient information to objectively appraise and either approve or disapprove of a proposed project. The EIR shall be forwarded to the developer who shall sign it and submit it to the Authority.

EIA report meaning: The EIA Report is a compilation of several important project components, including the project description, the assessment of its environmental and social impacts, mitigation measures, and related management and monitoring plans. During this step, all the information gathered during the previous steps is compiled into a comprehensive report that analyzes and synthesizes the data, structuring it as stipulated in the terms of reference (TOR).

Purpose of developing the EIA report: informational document" intended to inform public agency decision-makers and the public of the significant environmental effects of a project, identify possible ways to minimize the significant effects and describe reasonable alternatives to the project.

Advantages of developing an EIA report

Approach for developing an EIA report: The general approach to preparing an EIA report is to structure the information and interpret the material collected (such as research outcomes) in such a way that it provides reasoning for the suggested mitigation measures and changes in project implementation. The document structure and interpretation are guided by the Terms of Reference that the project is subjected to.

2.3.3. STRUCTURE OF INFORMATION IN EIA REPORT

Interpret the material collected

Structure of EIA report format: The main aim of an EIA Report is to provide prudent information for two types of audiences – decision-makers and people potentially affected by a Project. The Report, therefore, must communicate effectively with these audiences.

The quality should reflect the following:

- Table of contents
- Acronyms and abbreviations
- Executive summary
- General information
- Project and alternatives description
- Environmental setting
- Assessment of impacts
- Mitigation and monitoring measures
- Environmental management plan

- Commitment Statement
- Annexes

SUBMISSION OF EIA REPORT TO THE AUTHORITY

After a developer has reviewed the EIA report and, if necessary, written an addendum, these documents, which should be signed by the EIA experts, are submitted by the developer to REMA. The developer shall submit at least five copies of the EIA report to the Authority. When submitting EIA documents to the Authority, developers shall indicate any information, which they wish to remain confidential. All such confidential information shall only be privy to the developer, EIA experts and the Authority. The Authority shall ensure that for any project ready for review, three principal documents are available, namely:
i) Environmental Impact Report (EIR) including Environment Management Plan (EMP),
ii) Developer's Environmental Impact Report Addendum (where applicable), iii) Public Hearing Report.

2.3.4. EIA REPORT REVIEW AND DECISION-MAKING

Review of EIA documents submitted to the Authority enables subsequent decision-making on either approval or disapproval of a project. **a) Public Hearing**
RDB is responsible for conducting public hearings during the EIA process. The purpose of a public hearing is to furnish interested and affected parties and the public with an opportunity to comment on, or raise issues relevant to an application for environmental authorization.
Participants to the public hearing will include:
Government agencies with responsible for licensing (RDB, REMA, RNRA, RBS,MINICOM, MINIRENA, MININFRA, MIFOTRA), concerned Local Authorities; Private sector Federation; National Police Fire Brigade; Professional Associations including Impact Assessors and Engineers' Body (if and when registered); Local community representatives; non-governmental organizations and the developer.
After verification of the EIA study report, the assigned focal person may have comments or request for additional information.

In this case the applicant will be notified via email and SMS and he / she will be requested to send any additional information through the online application system.

For big projects, it might take more than 20 days after the submission of the EIA report to obtain an EIA Certificate. In this case, the client will be informed at the time of submission of the EIA report. The assigned focal person will call the applicant to let him/her know when the Environmental Impact Assessment (EIA) Certificate is ready.

LEARNING UNIT 3 - CONTROL HUMAN PROJECTS IMPACT ON ENVIRONMENT

Learning Outcomes:

3.1 Adapt the project to the environment for its management

3.2 Control environmental pollution

3.3 Apply climate change mitigation measures

LEARNING OUTCOME 3.1: ADAPT THE PLANNED PROJECT TO THE ENVIRONMENT FOR ITS MANAGEMENT

 Dimensions of the environment Components (physical and socio-economic) Scale/Space Time

✓ **Nature of impact:** Environmental impacts are changes in the natural or built environment, resulting directly from an activity that can have adverse effects on the air, land, water, fish, and wildlife or the inhabitants of the ecosystem.

✓ **Ways to protect the environment**

 Reduce, reuse, and recycle

A plastic bottle is estimated to take about 1,000 years to decompose (or 450 years if it's left outside). Consumers play an important role in reducing pollution. Following the "three **R**s" (**reduce, reuse, and recycle**) is a way to do it.

- **Reduce**: we can take this primary step when using water, electricity or gas (e.g., switching off devices or turning off the tap when we're not using them), or even when selecting items at the supermarket (e.g., products from close to where we live waste less on transport).

- **Recycle:** we can do this simple act by sorting waste into specific containers. Processing recycled waste properly reduces gases and toxic waste and leads to newly manufactured products.
- **Reuse**: this is when we repurpose things for a "second life", like turning glass food jars into pen holders or old clothes into dust rags.

- **Plant trees:** Trees provide food and oxygen. They help save energy, clean the air, and help combat climate change.
- **Consume less:** Overconsumption means consuming resources that we cannot replenish or that cannot sustain themselves at the rate we are consuming them. Ecosystems are unable to cope with excessive resource extraction, resulting in biodiversity loss and the deterioration of the natural world
- **Compost:** Organic waste in landfills generates, methane, a potent greenhouse gas. By composting wasted food and other organics, methane emissions are significantly reduced. Compost reduces and, in some cases, eliminates the need for chemical fertilizers.
- **Upcycle more:** Upcycling reduces the number of materials and products in landfills. It also minimizes the use of natural resources and raw materials. For that reason, upcycling leaves a smaller footprint reduces consumption and minimizes the strain on valuable resources, such as fuel, forests, and water supplies.
- **Use fewer chemicals:** Choose non-toxic chemicals in the home and office.
- **Walk, bike, or carpool:** Drive less.
- **Conserve electricity: turning off lights**: Turn them off when you're not in a room or if there's natural light. Use LED bulbs to achieve optimum lighting at a lower cost, unplugging devices when they are not using: The so-called "standby power" makes up a significant portion of your monthly bill.
- **Use long-lasting light bulbs:** Energy-efficient light bulbs reduce greenhouse gas emissions. Also, flip the light switch off when you leave the room!
- **Turn off your device**
- **Use a natural sunscreen**

- **Choose sustainably:** Sustainability is important for many reasons including Environmental Quality – In order to have healthy communities, we need clean air, natural resources, and a non-toxic environment
- **Spend time in your garden:** Home gardens also provide a number of ecosystem services such as habitats for animals and other beneficial organisms, nutrient recycling, reduced soil erosion, and enhanced pollination
- **Conserve water: T**he less water you use, the less runoff and wastewater that eventually end up in the ocean.

LEARNING OUTCOME 3.2: CONTROL ENVIRONMENTAL POLLUTION

✓ **Pollution**

- **Meaning:** Pollution is the introduction of harmful materials into the environment. These harmful materials are called pollutants. Pollutants can be natural, such as volcanic ash. They can also be created by human activity, such as trash or runoff produced by factories. Pollutants damage the quality of air, water, and land.
  - **Disadvantages:** Pollution has a negative impact on the environment in that it causes significant harm to the natural world. Pollution has an impact on the land, water, air, and nearby areas. Nature has given us a gift in the form of our environment, but as it is continuously depleting, the natural cover or protection is getting lost. The ozone layer, which shields us from the sun's harmful ultraviolet rays, has decreased as a result of deforestation, and pollution has compounded this problem by hastening the ozone layer's deterioration.

✓ **Types of environmental pollution:**

- **Air pollution:** Air pollution refers to the release of harmful contaminants (chemicals, toxic gases, particulates, biological molecules, etc.) into the earth's atmosphere. These contaminants are quite detrimental and, in some cases, pose serious health issues. Some cause that contributes to air pollution are:
  - Burning fossil fuels
  - Mining operations
  - Exhaust gases from industries and factories

The effects of air pollution vary based on the kind of pollutant. But generally, the impact of air pollution ranges from:

- Increased risk of respiratory illness and cardiovascular problems
- Increased risk of skin diseases
- May increase the risk of cancer
- Global warming
- Acid rain
- Ozone depletion
- Hazards to Wildlife

- **Water pollution:** Water pollution is said to occur when toxic pollutants and particulate matter are introduced into water bodies such as lakes, rivers, and seas. These contaminants are generally introduced by human activities like improper sewage treatment and oil spills. However, even natural processes such as eutrophication can cause water pollution.
  - Other significant causes of water pollution include:
  - Dumping solid wastes in water bodies
  - Disposing of untreated industrial sewage into water bodies
  - Human and animal wastes
  - Agricultural runoff containing pesticides and fertilizers
- **Noise pollution:** Noise pollution refers to the excessive amount of noise in the surrounding that disrupts the natural balance. Usually, it is man-made, though certain natural calamities like volcanoes can contribute to noise pollution.

In general, any sound which is over 85 decibels is considered to be detrimental. Also, the duration an individual is exposed plays an impact on their health. For perspective, a normal conversation is around 60 decibels, and a jet taking off is around 15o decibels. Consequently, noise pollution is more obvious than other types of pollution.

Noise pollution has several contributors, which include:

- Industry-oriented noises such as heavy machines, mills, factories, etc.
- Transportation noises from vehicles, airplanes, etc.
- Construction noises
- Noise from social events (loudspeakers, firecrackers, etc.)
- Household noises (such as mixers, TV, washing machines, etc.)

- **Thermal pollution:** Thermal pollution may also be referred to as thermal enrichment. When the temperature of a natural body of water suddenly increases or decreases, thermal pollution occurs. Industrial machinery and power plants are big contributors to thermal pollution. An example of thermal pollution is when industrial sites and power plants often take water from a natural source. When the water is returned, the temperature has been altered. In our modern society, thermal pollution is a genuine problem. One of the significant effects of thermal pollution is a change in oxygen levels, destroying ecosystems and communities.

Causes of thermal pollution:

1. Use of water as a cooling agent in industrial plants
2. Soil erosion
3. Runoff from paved surfaces Runoff from roads, parking lots, and other surfaces can affect water temperature.
4. Natural causes Humans are not the only cause of thermal pollution.
5. Deforestation

- **Land pollution:** Many of the same pollutants that foul the water also harm the land. Mining sometimes leaves the soil contaminated with dangerous chemicals. Pesticides and fertilizers from agricultural fields are blown by the wind. They can harm plants, animals, and sometimes people. Some fruits and vegetables absorb the pesticides that help them grow. When people consume fruits and vegetables, pesticides enter their bodies. Some pesticides can cause cancer and other diseases
- **Radioactive pollution:** Radioactive contamination is defined as the deposition or introduction of radioactive substances into the environment, where their presence is unintended, or the levels of radioactivity are undesirable. Such type of pollution is harmful to life due to the emission of ionizing radiation. This type of radiation is potent enough to cause damage to tissues and DNA in genes.

Radioactivity can occur in one of two ways:

- Naturally occurring radioactivity
- Man-made radioactivity

Natural radioactivity, as the name suggests, occurs naturally in our environment. Some radioactive elements such as uranium and thorium are present in rocks and soil, albeit in trace

quantities. Interestingly, humans and all other living organisms contain nuclides such as carbon14, which are created by cosmic rays.

Man-made radioactivity is the result of a nuclear weapon discharge or a nuclear reactor containment breach. In such scenarios, all living organisms in the vicinity of the nuclear event will become contaminated by fission products and remnants of nuclear fuel. This can be in the form of radioactive dust or even particles that are found on various surfaces.

- ✓ **Soil pollution:** Soil pollution, also called soil contamination, refers to the degradation of land due to the presence of chemicals or other man-made substances in the soil. The xenobiotic substances alter the natural composition of soil and affect it negatively. These can drastically impact life directly or indirectly. For instance, any toxic chemicals present in the soil will get absorbed by the plants. Since plants are producers in an environment, it gets passed up through the food chain. Compared to the other types of pollution, the effects of soil pollution are a little more obscure, but their implications are very noticeable.

Some of the common causes of soil pollution are:

- ➢ Improper industrial waste disposal
- ➢ Oil Spills
- ➢ Acid rain which is caused by air pollution
- ➢ Mining activities
- ➢ Intensive farming and agrochemicals (like fertilizers and pesticides) ➢ Industrial accidents

- ✓ **Ways to control environmental pollution:**

Air pollution: There are many small, but critical sources of air pollution in our homes and neighborhoods. Such sources — vehicles, construction equipment, lawnmowers, dry cleaners, backyard fires, and auto-body shops — are located where we live and work. Total emissions from these smaller but widespread sources are significantly greater than all the industrial sources in the state combined. The following are the ways of controlling air pollution

- ➢ Drive your car less. ...
- ➢ Keep your car in good repair. ...
- ➢ Turn off your engine. ...
- ➢ Don't burn your garbage. ...
- ➢ Limit backyards fire in the city. ...

- Plant and care for trees. ...
- Switch to electric or hand-powered lawn equipment. ...
- Use less energy.

Preventive technique

Gaseous criteria pollutants, as well as volatile organic compounds (VOCs) and other gaseous air toxics, are controlled by means of three basic techniques: absorption, adsorption, and incineration (or combustion). These techniques can be employed singly or in combination.

✓ **Water pollution treatment:** Water treatment removes contaminants and undesirable components, or reduces their concentration so that the water becomes fit for its desired enduse. This treatment is crucial to human health and allows humans to benefit from both drinking and irrigation use.

- **Physical treatment:** Physical water treatment typically consists of filtration techniques that involve the use of screens, sand filtration or cross-flow filtration membranes.
  Screens: Typically used as a pretreatment method to remove larger suspended material.
  Sand and/or Multimedia Filtration: Frequently used to filter suspended solids.
- **Chemical treatment:** Chemical water treatment can be successfully used for the removal of both toxic organic and inorganic compounds, and physical–chemical methods are intended to remove a wide range of contaminants from water starting from organic and inorganic compounds and ending with biological pollution.
- **Biological treatment:** Biological wastewater treatment is designed to degrade pollutants dissolved in effluents by the action of microorganisms. Microorganisms utilize these substances to live and reproduce. Pollutants are used as nutrients.

✓ **Noise pollution**

- **Silencers:** Absorptive Silencers for Noise Control
  These ventilation silencers use porous or fibrous materials that absorb sound waves and transfer sound energy into heat energy through friction. This weakens the sound waves to reduce noise in your facility. Absorptive ventilation silencers are best for higherfrequency noises.
- **Noise absorbing:** porous materials obtained from synthetic fibers, such as mineral wool or glass wool, are commonly used for thermal insulation and sound absorption, because

of their high performance and low cost. Their diffuse-field sound absorption coefficient is very high at mid-high frequencies.

- **Muffs and earplugs:** Properly fitted earplugs or muffs reduce noise 15 to 30 dB. The better earplugs and muffs are approximately equal in sound reductions, although earplugs are better for low-frequency noise, and earmuffs for high-frequency noise**.**
- **Soundproof windows:** "Soundproof" is shorthand for noise reduction windows that block up to 90% to 95% of noise coming through windows. Sounds with low frequencies, like garbage trucks grinding trash, are harder to block than sounds with higher frequencies like birds chirping**.**
- **Planting trees:** Trees are able to reduce or attenuate sound by intercepting sound waves and changing their behavior. Different plant parts reduce noise by absorbing, deflecting, or refracting sound waves depending on their physical characteristics.

✓ **Thermal pollution**

- **Cooling ponds**: Cooling ponds or reservoirs are the simplest methods of controlling thermal discharges. In cooling ponds, heated effluents on the surface of water maximize the dissipation of heat into the atmosphere and minimize the area and volume of water. This is the simplest and cheapest method that cools the water to a considerably low temperature. However, the method alone is less desirable as well as inefficient in terms of air-water contact.
- **Cooling towers:** After using water from water sources for cooling purposes, it is subsequently returned to the water body after passing through the condenser, which is termed the cooling process.

  Therefore, cooling towers are designed to control the temperature of water to make the cooling process more effective. Cooling towers are mainly used to dissipate the recovered waste heat to eliminate the problems of thermal pollution.
- **Spray ponds:** Cooling is accomplished through heat exchange with the surrounding air. This includes both conductive heat transfer between the water droplets and the surrounding air as well as evaporative cooling which provides by far the greatest portion, typically 85 to 90 percent, of the total cooling.

- **Artificial lake:** Artificial lakes are man-made water bodies that offer a possible alternative. The heated effluents may be discharged into the lake at one end, and the water may be withdrawn from the other end for cooling purposes. The heat is eventually dissipated through evaporation. However, these lakes have to be rejuvenated continuously.

✓ **Prevention of Land Pollution**

There are many ways that can be taken on a bigger level as well as a small one. It can help in preventing land pollution. In order to execute it, the government and citizens must come together to do so. We can begin with:

- **Reducing the usage of chemicals and pesticides**

Reducing the use of chemical fertilizers and pesticides can help in preventing land pollution. In addition to contaminating the soil, they also do no good to the crops.

Therefore, farmers must use natural ingredients instead of these harmful toxins. They can switch to manure and bio-fertilizers. This will help in minimizing land pollution**.**

- **Reforestation**

As we know that forests and grassland bind the soil. We must practice reforestation to protect the land from pollution. It will further prevent floods and soil erosion. Moreover, it will make the land more fertile and enhance biodiversity.

- **Recovering and Recycling Material**

We must recycle waste material to stop it from polluting the land. We must properly dispose of organic and inorganic waste separately. It will help in lessening the burden of landfills.

Moreover, it will save more natural resources.

✓ **Radioactive pollution prevention**

Activities involving radioactive materials, such as mining, handling and processing radioactive materials, handling and storage of radioactive waste, and the use of radioactive reactions to generate energy (nuclear power plants), as well as the use of radiation in medicine (e.g. X-rays) and research, are all examples of human activities that can release radiation.

➢ **Measures to Control Radioactive Pollution**

1. Proper Storage and Disposal of Radioactive Waste

Radiation can still be found in radioactive waste. As a result, it cannot be disposed of in the same manner as regular waste.

Storage**:** It cannot be buried or incinerated. Since seepage is a possibility, this waste should be stored in large, thick concrete containers.

Dilution: Since storage may not be practicable, another alternative is to dilute the radiation. ***Professional Assistance***: Professional assistance should always be sought because there are no straightforward ways to dispose of radioactive waste.

Low-level Waste: Low-level waste disposal is simple and may be done in practically any place in a safe manner.

Used Fuel: Used fuel is typically stored for at least five years underwater and subsequently in dry storage.

Deep Geological Disposal: The greatest solution for the eventual disposal of the most radioactive waste produced is deep geological disposal, which is widely accepted.

2. Proper Labeling

- Any material containing radioactive material must be labeled, with the required precautions stated on the label's content.
- As waste is added, information such as permit number, date, isotope, and radioactivity must be written legibly and precisely on the accompanying tag.
- The presence of organic or aqueous solvents, complexing or chelating chemicals (e.g., EDTA) in the liquid waste supplied to the container must be stated on the waste tag.
- The reason for this is that even light contact with radioactive material can allow radiation to enter the body.
- Containers containing such materials should be encouraged for the use of protective equipment when handling them.

3. Radiation Exposure Protection

- Distance, time to exposure, and shielding are the three concepts of radiation protection.
- The exposure diminishes as the distance from the source increases squared.
- To complete a certain activity, the exposure duration should be kept as short as possible.

- The total exposure should be kept below the limit dose in the case of occupational needs.

4. Radiation Contamination Protection

Inhaling radioactive particles is especially dangerous.

They continue to irradiate the body until they are expelled.

Using laboratory hoods, air filters, and exhaust systems, eliminating dry sweeping, wearing protective clothing and radiation indicators (to see the total amount of radiation to which one has been exposed), wearing respirators when indicated, and prohibiting smoking and eating where radioactive material is used can all help to reduce this risk.

5. Collection, storage on disposal

Most radioactive wastes are managed in the following way:

- **Planning and preparation**: Wherever possible, sites aim to reduce the amount of waste they produce. They also plan how to manage waste before it arises.
- **Waste treatment**: Waste treatment typically occurs in some form soon after it arises. Waste treatment techniques depend on the type of waste and the intended disposal route. Examples of treatment include decontaminating, shredding, compacting, drying and solidifying the waste.
- **Packaging**: Most radioactive waste requires packaging in specially engineered containers for safe storage and disposal. This also allows for easier handling and transport.
- **Storage**: Interim facilities will store certain waste types until a suitable disposal route becomes available. Storage may last from a few months to many decades.
- **Disposal**: This involves placing wastes into engineered facilities where they will remain permanently.

LEARNING OUTCOME 3.3: APPLY CLIMATE CHANGE MITIGATION MEASURES

Climate change is a long-term change in the average weather patterns that have come to define Earth's local, regional and global climates.

3.3.1. Climate change indicators

Temperatures are rising, snow and rainfall patterns are shifting, and more extreme climate events like heavy rainstorms and record-high temperatures – are becoming more common.

Many of these observed climate changes are linked to rising levels of greenhouse gases in our atmosphere, caused by human activities.

Surface temperature: According to the 2022 Global Climate Report from NOAA National Centers for Environmental Information, every month of 2022 ranked among the ten warmest for that month, despite the cooling influence of the La Niña climate pattern in the tropical Pacific. The "coolest" month was November, which was 1.35 ˚F (0.75 ˚C) warmer than average.

Past and future changes in global temperature

Though warming has not been uniform across the planet, the upward trend in the globally averaged temperature shows that more areas are warming than cooling. According to NOAA's 2021 Annual Climate Report, the combined land and ocean temperature has increased at an average rate of 0.14 degrees Fahrenheit (0.08 degrees Celsius) per decade since 1880; however, the average rate of increase since 1981 has been more than twice as fast: 0.32 °F (0.18 °C) per decade.

The amount of future warming Earth will experience depends on how much carbon dioxide and other greenhouse gases we emit in the coming decades. Today, our activities—burning fossil fuels and clearing forests—add about 11 billion metric tons of carbon (equivalent to a little over 40 billion metric tons of carbon dioxide) to the atmosphere each year. Because that is more carbon than natural processes can remove, atmospheric carbon dioxide increases each year.

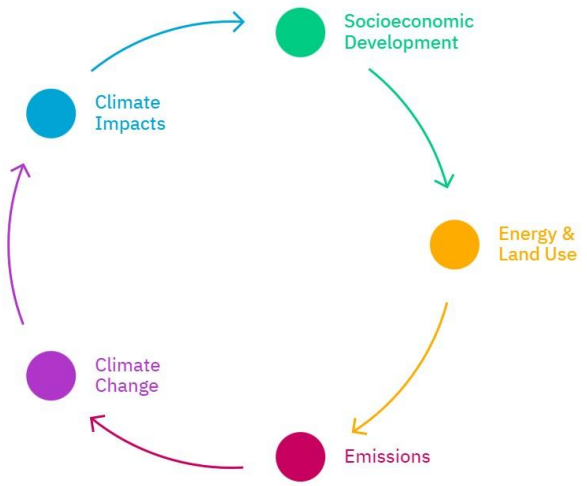
Greenhouse gases (GHGs) and Ozone Layer: The depletion of the ozone layer is caused mainly by the increase in emissions of chlorine- and bromine-containing compounds like CFCs, halons, carbon tetrachloride, methyl chloroform, and methyl bromide. Emissions of greenhouse gases can affect the depletion of the ozone layer through atmospheric interaction.

We studied the interactions in the atmosphere between the greenhouse effect and stratospheric ozone depletion from the point of view of past and future emissions of the anthropogenic compounds: CFCs, halons, CH4, N2O, NOx, CO, and CO2. In our investigation, the increase in emissions of chlorine- and bromine-containing compounds, largely responsible for the depletion of stratospheric ozone at mid-latitudes, was found to be -5.8% per decade from 1980 to 1990.

The socioeconomic implication of climate change

As the climate continues to change, millions of poor people face increasing challenges in terms of extreme events, health effects, food, water, and livelihood security, migration and forced displacement, loss of cultural identity, and other related risks.

Climate change and socioeconomic development are deeply intertwined. Social and economic activities are the main driver of climate change. In turn, climate change will have serious impacts on these activities, e.g., by rising sea levels and exposure to severe weather events.



Natural disasters and human mobility

With increasing global surface temperatures the possibility of more droughts and increased intensity of storms will likely occur. As more water vapor is evaporated into the atmosphere it becomes fuel for more powerful storms to develop. More heat in the atmosphere and warmer ocean surface temperatures can lead to increased wind speeds in tropical storms. Rising sea levels expose higher locations not usually subjected to the power of the sea and to the erosive forces of waves and currents.

Climatic-related human mobility can amplify factors contributing to communal unrest and undermining social cohesion. Competition over natural resources may exacerbate pressures that contribute to tension and may interact with the movements of people. In recent decades, droughts and other climatic stressors have afflicted areas already torn by strife, contributing to large-scale movements, food insecurity, and humanitarian situations. The return depends upon establishing peaceful conditions.

3.3.2. SOME OF THE CLIMATE MITIGATIONS

Erecting buildings and infrastructure that is safer and more sustainable.

Replanting forests and restoring damaged ecosystems. Diversifying crops so that they are better able to adapt to changing climates.

Investigating and developing innovative solutions to prevent and manage natural catastrophes.

Environmental and health impact

Mitigation technology and practices by sector

Energy: Mitigating climate change is about reducing the release of greenhouse gas emissions that are warming our planet. Mitigation strategies include retrofitting buildings to make them more energy efficient; adopting renewable energy sources like solar, wind, and small hydro; helping cities develop more sustainable transport such as bus rapid transit, electric vehicles, and biofuels; and promoting more sustainable uses of land and forests. ***In Rwanda***

- Establishment of new grid-connected renewable electricity generation capacity in the form of large-scale hydropower plants and solar PV power
- Installation of solar PV mini-grids in rural Communities
- Increase energy efficiency through demand-side measures and grid-loss reduction
- Promote environmentally sustainable use of biomass fuels

Transport: Promotion of public transport, improvement of transport infrastructure, setting vehicles' emission standards and regulations, and integrated national transportation planning

Buildings:

Mitigation strategies for climate change in buildings are mainly focused on promoting energy saving, the use of renewable energies, the proper management of waste, the integration of vegetation in building projects (such as roofs, walls, and green terraces), and incorporation of elements that facilitate the use of non-motorized transport (such as the installation of parking lots for bicycles or charging stations for electric vehicles), among others, and are applicable in different measures both in existing buildings and in new buildings.

Green building is a solution to the mitigation of climate change. Energy-efficient construction for new buildings, and renovation of existing buildings, can reduce 40-50% of the energy used for space and water heating in buildings. However, if current inefficient building practices continue, buildings could account for 70% of CO2 emissions by 2050. Life Cycle Assessment (LCA),

carbon-focused, building rating systems are “material neutral” but green building assessment schemes without a scientific LCA basis underestimate the carbon intensity issue. Some countries have green building rating systems, and other countries are considering establishing them, but existing green building rating systems lack an LCA basis.

Industry:

- Scale up resource efficiency technologies to reduce energy demand in agro-processing industries
- Establishment of an Eco-industrial park or Green Industry Complex **Agriculture:**
- Mainstreaming agroecology techniques using spatial plant stacking as in agroforestry, kitchen gardens, nutrient recycling, and water conservation to maximize sustainable food production
- Utilizing resource recovery and reuse through organic waste composting and wastewater irrigation
- Using fertilizer-enriched compost
- Soil conservation and land husbandry **Forestry**
- Promote afforestation/reforestation of designated areas through enhanced germplasm and technical practices in planting and post-planting processes
- Employ Improved Forest Management for degraded forest resources

Waste

The result of the throwaway society is large amounts of materials found in landfill, which can be considered as a new source to mine resources from.

The exploration of possible solutions to eliminate waste in our society has given rise to ideas developed as part of the circular economy. By redesigning products and processes, the waste from one product can become the raw material of another.

Waste minimization is a process of elimination that involves reducing the amount of waste produced in society and helps eliminate the generation of harmful and persistent wastes, supporting the efforts to promote a more sustainable society. Waste minimization involves redesigning products and/or changing societal patterns, concerning the consumption and production, of waste generation, to prevent the creation of waste.

Waste management focuses on processing waste after it is created, concentrating on reuse, recycling, and waste-to-energy conversion rather than eliminating the creation of waste in the initial phases of production. Waste minimization involves efforts to minimize resource and energy use during manufacture. For the same commercial output, usually the less materials are used, the less waste is produced. Waste minimization usually requires knowledge of the production process, cradle to grave analysis (the tracking of materials from their extraction to their return to earth) and detailed knowledge of the composition of the waste.

3.3.3. CLIMATE CHANGE MITIGATION MEASURES

In general, we all have to work together to mitigate climate change effects through environmental friendly practices:

- Retrofitting buildings to make them more energy sufficient
- Adopting renewable energy use such as solar, wind, water
- Helping cities develop more sustainable transport such as bus rapid transit, electric vehicles, and use of biofuels
- Promoting sustainable land use and forests
- Encourage recycle of used materials
- Plant more and more trees and protect well natural resources

3.3.3.1 SOME OF THE CLIMATE CHANGE MITIGATION MEASURES

I. Use of renewable energies

The first way to prevent climate change is to move away from fossil fuels. What are the alternatives? Renewable energies like solar, wind, biomass and geothermal.

II. Energy & water efficiency

Producing clean energy is essential, but reducing our consumption of energy and water by using more efficient devices (e.g. LED light bulbs, innovative shower systems) is less costly and equally important.

II. Encourage sustainable transportation

Promoting public transportation, carpooling, but also electric and hydrogen mobility, can definitely help reduce CO_2 emissions and thus fight global warming.

III. Sustainable infrastructure

In order to reduce the CO_2 emissions from buildings - caused by heating, air conditioning, hot water or lighting - it is necessary both to build new low energy buildings, and to renovate the existing constructions.

IV. Sustainable agriculture & forest management

Encouraging better use of natural resources, stopping massive deforestation as well as making agriculture greener and more efficient should also be a priority.

V. Responsible consumption & recycling

Adopting responsible consumption habits is crucial, be it regarding food (particularly meat), clothing, and cosmetics or cleaning products. Last but not least, recycling is an absolute necessity for dealing with waste.

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